

Finding the Sweet Spot: An Empirical Study on Dataset Size, Performance, and Efficiency in Relation Extraction

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Research Objective

Investigate how dataset size influences relation extraction performance.

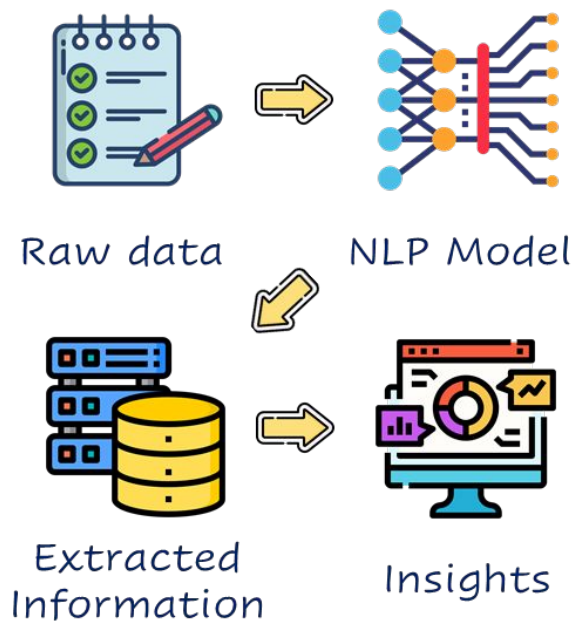
Why it matters?

**Cost-Effectiveness
of Data Annotation**

**Model Generalization
and Overfitting**

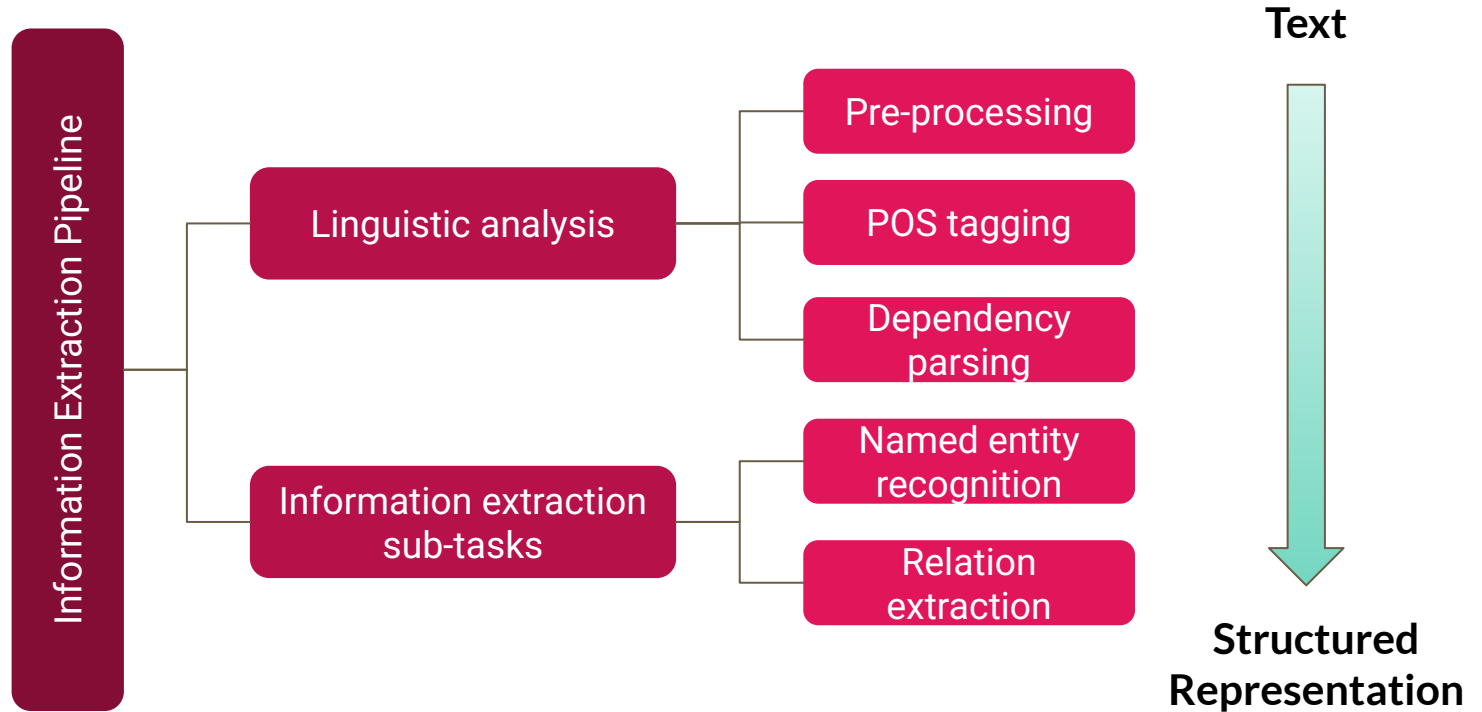
**Performance Gains
and Efficiency**

Introduction



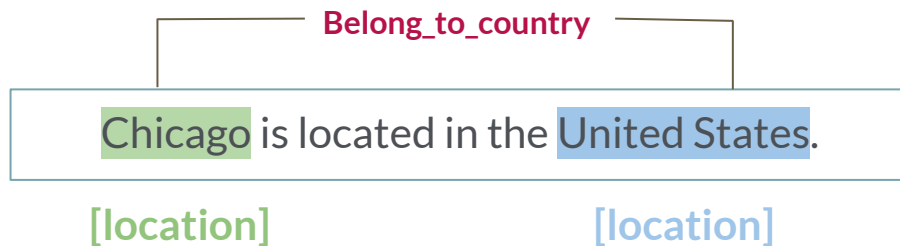
[<https://devblogs.microsoft.com/cse/tag/named-entity-recognition/>]

Information Extraction



Relation Extraction

The process of predicting and labeling semantic relationships between named entities.



Relation Extraction - cont.

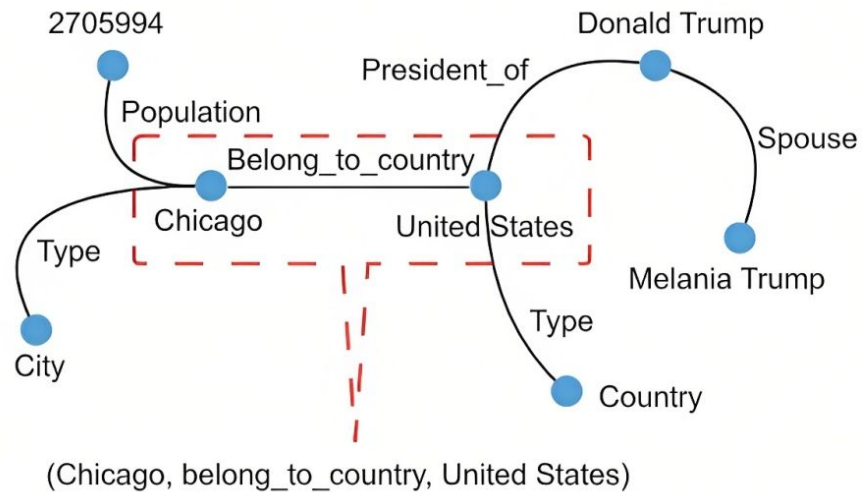
Unstructured Text Data

Chicago is located in the United States.

Head entity

Tail entity

Structured Representation



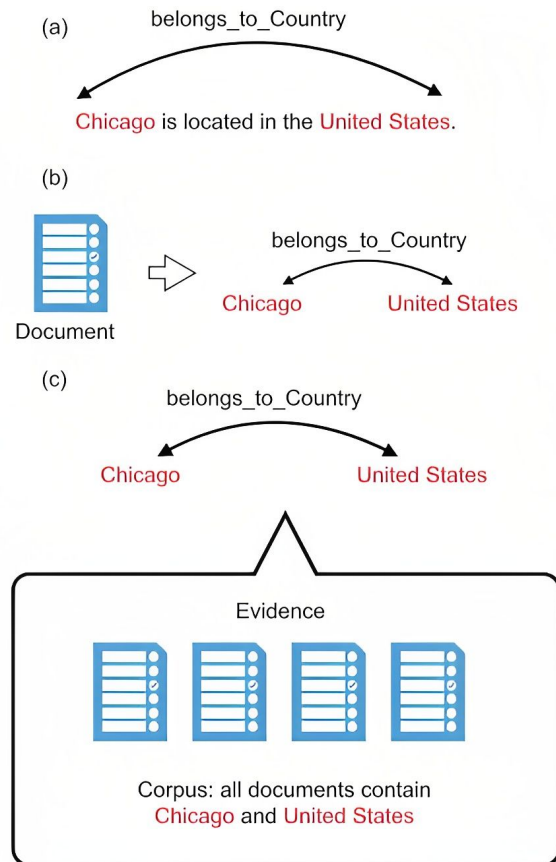
[Liu Kang - 2020]

Relation Extraction - cont.

(a) Sentence-level

(b) Document-level

(c) Corpus-level



[Liu Kang - 2020]

Relation Extraction - cont.

- Supervised methods
 - Consider RE as a multi-class classification problem.
- Distant supervision methods (weakly supervised)
 - Aligning relational triplets from a knowledge base with free text to automatically generate labeled training data.
- Semi-supervised methods
 - Methods in this setting typically rely on **bootstrapping**, or **bootstrap learning**.
- Unsupervised
 - The context information of different entity pairs with the same semantic relation is relatively similar.

Related Work

Relation Extraction using Attention-Based Neural Networks

- Attention-Based Convolutional Neural Networks for Semantic Relation Extraction (**Att-CNN**).

[Yatian Shen, Xuanjing Huang]

- Attention-Based Bidirectional Long Short-Term Memory Networks for Relation Classification (**Att-BLSTM**).

[Peng Zhou, et. al]

Relation Extraction using Pre-trained Language Models

- Enriching Pre-trained Language Model with Entity Information for Relation Classification (**R-BERT**).

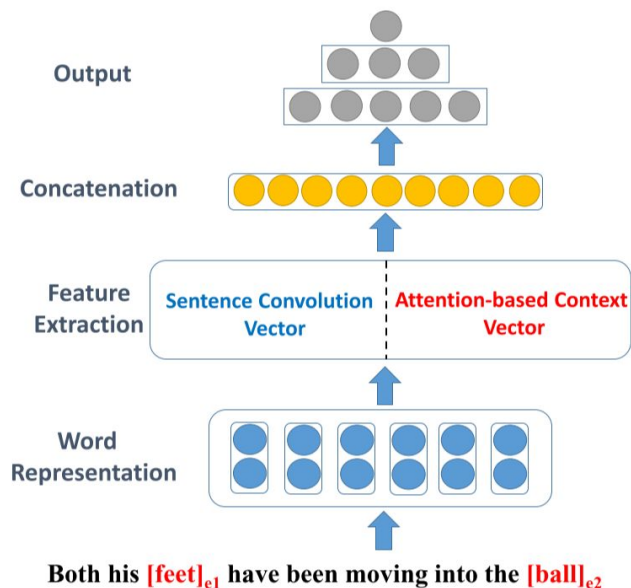
[Wu and He]

- Matching the Blanks: Distributional Similarity for Relation Learning (**MTB**).

[Soares et.al]

Related Work

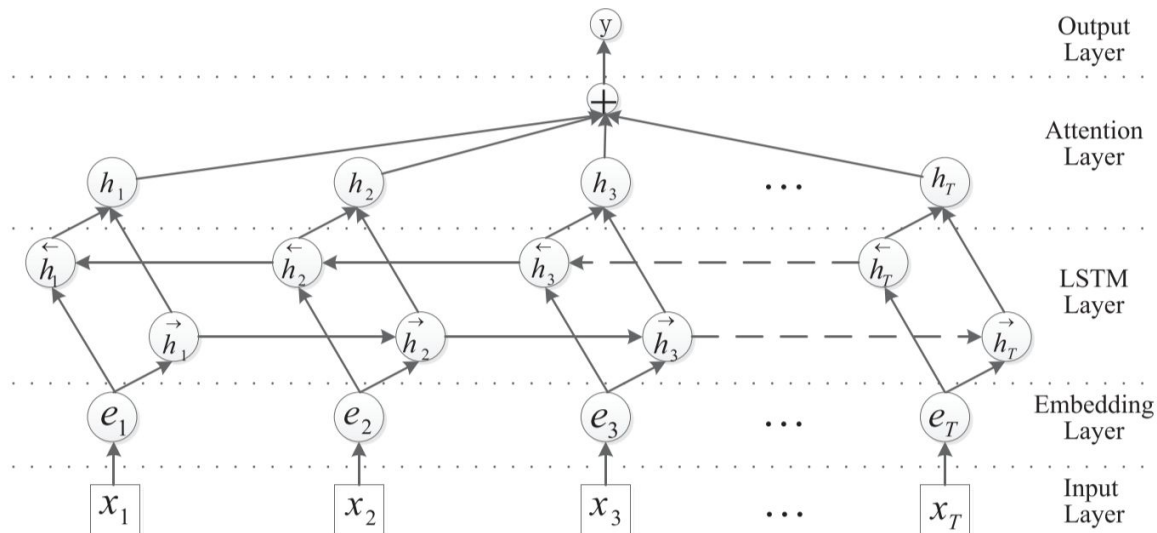
Attention-Based Convolutional Neural Networks for Semantic Relation Extraction



[Yatian Shen, Xuanjing Huang]

Related Work

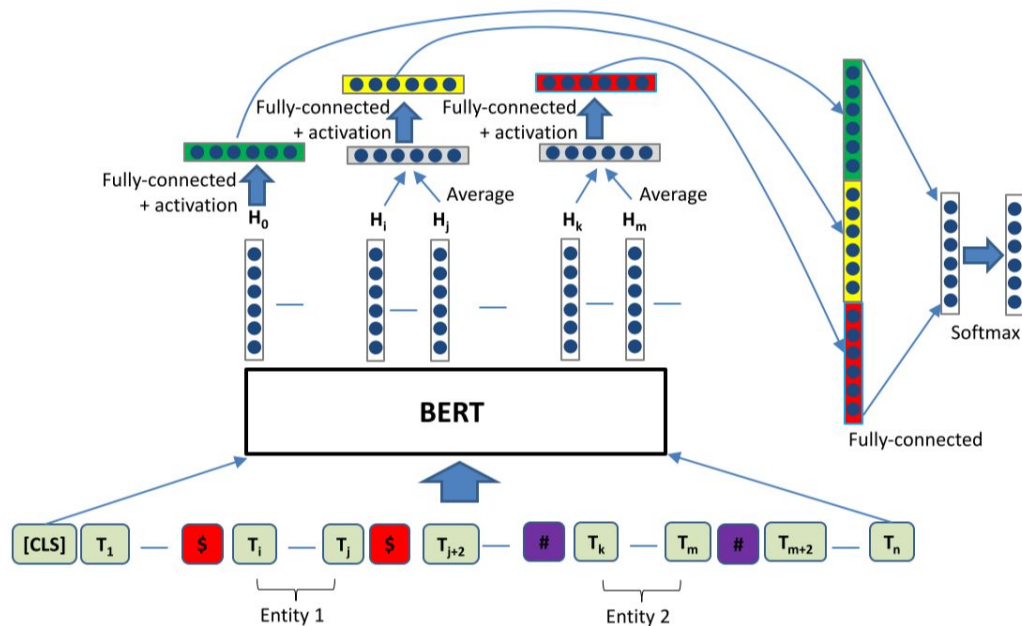
Attention-Based Bidirectional Long Short-Term Memory Networks for Relation Classification



[Peng Zhou, et. al]

Related Work

Enriching Pre-trained Language Model with Entity Information for Relation Classification (R-BERT)



[Wu and He - 2019]

Related Work

Matching the Blanks: Distributional Similarity for Relation Learning (MTB)

“If two entities participate in a relation, any sentence that contain those two entities might express that relation.” [Mintz et al. - 2009]

Russian forces had taken control of major cities in **Ukraine**'s east.

Russian troops crossed **Ukraine**'s border bring with them hundreds of tanks.

The city's **British** rulers and **Russian** refugees may be long gone but their food has remained



Dataset

Dataset	Examples	Neg. examples(%)	Relations	Supervision
SemEval-2010 Task 8	10,717	17.4%	19	traditional

Datasets - cont.

SemEval-2010 Task 8

The dataset for the SemEval-2010 Task 8 is a dataset for classification of semantic relations between pairs of nominals.

"He had chest pains and <e1>headaches</e1> from <e2>mold</e2> in the bedrooms."

Cause-Effect(e2,e1)

Contribution

Examining how the size of a training dataset affects the performance of a relation classification task.

Goal: Impact of dataset size on relation classification models.

Approach: Perform 4 different Relation Classification models over 8 subsets of SemEval-2010 Task 8 Dataset.

Results: Impact of training dataset size on the performance of models reduce as much as the size of the training data increase regardless of what kind of classification model is used.

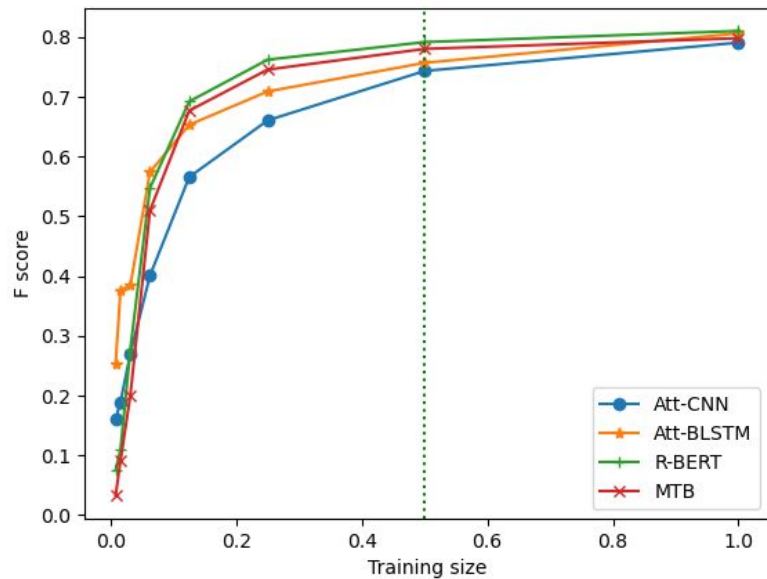
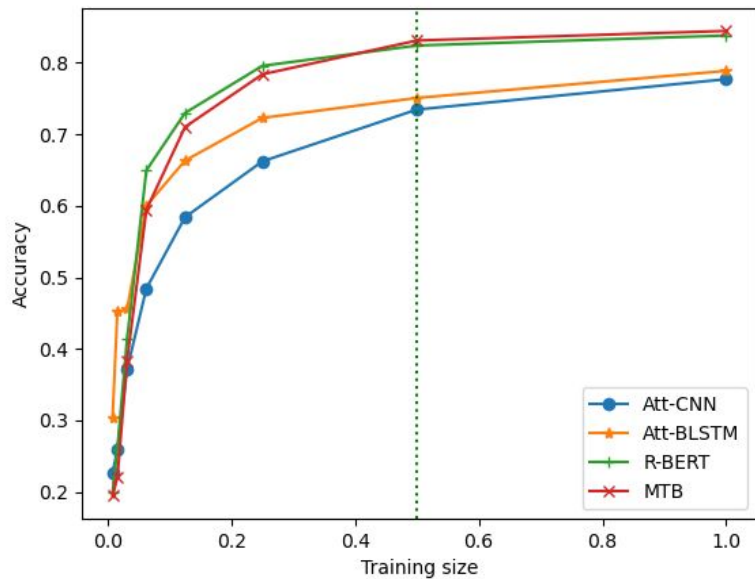
Contribution - cont.

SemEval-2010 Task 8 Subsets

Subset	Portion	Number of Samples
1	1/128	62
2	1/64	125
3	1/32	250
4	1/16	500
5	1/8	1000
6	1/4	2000
7	1/2	4000
8	1	8000

Contribution - cont.

Experiments

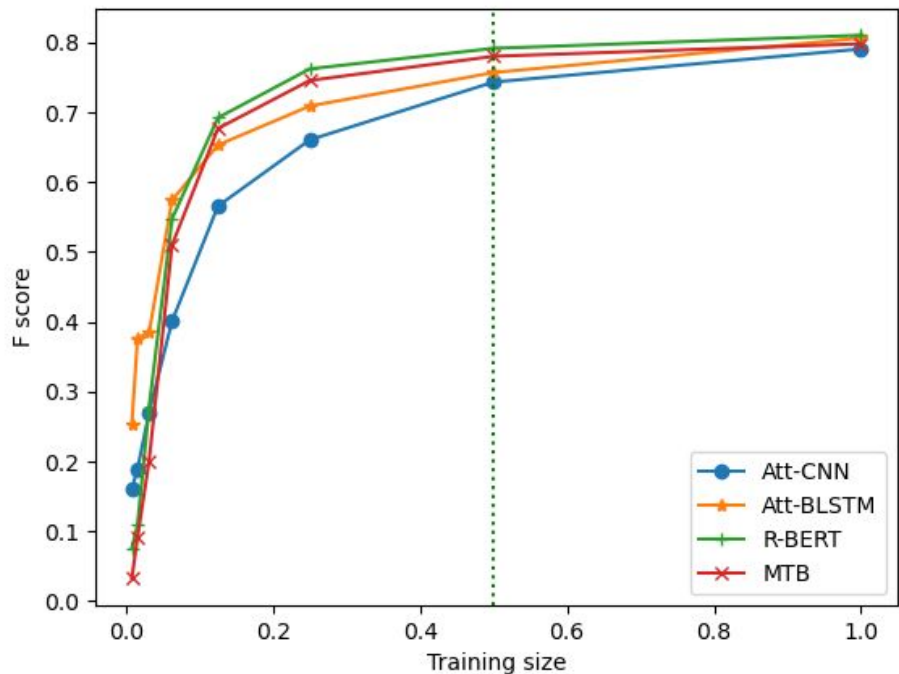


Contribution - cont.

Findings

Diminishing Returns:

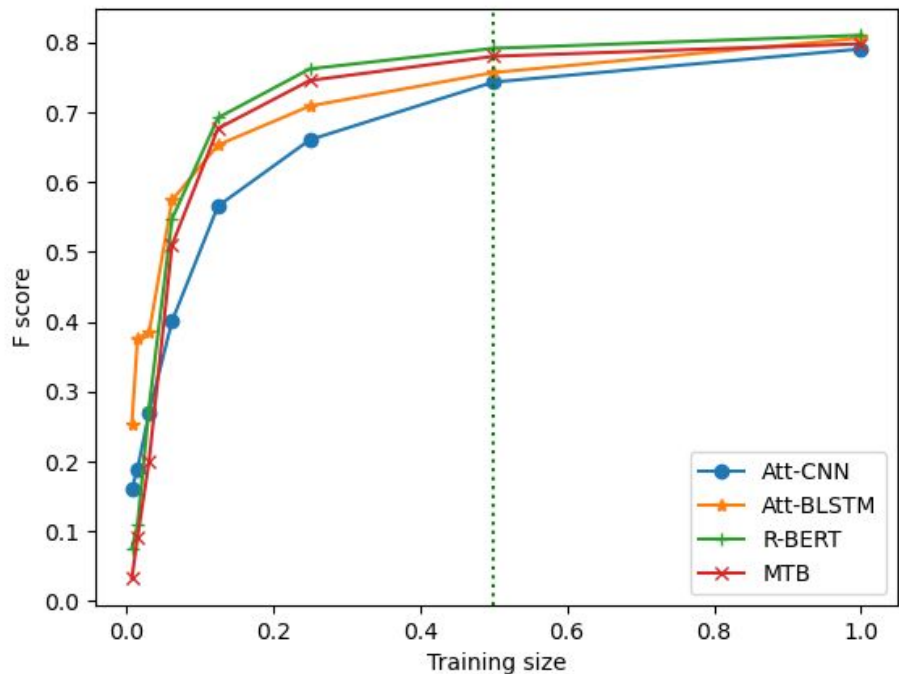
Performance improves significantly with increasing data but stabilizes **beyond 50%** of the full dataset—indicating that further annotation may not be cost-effective.



Contribution - cont.

Findings

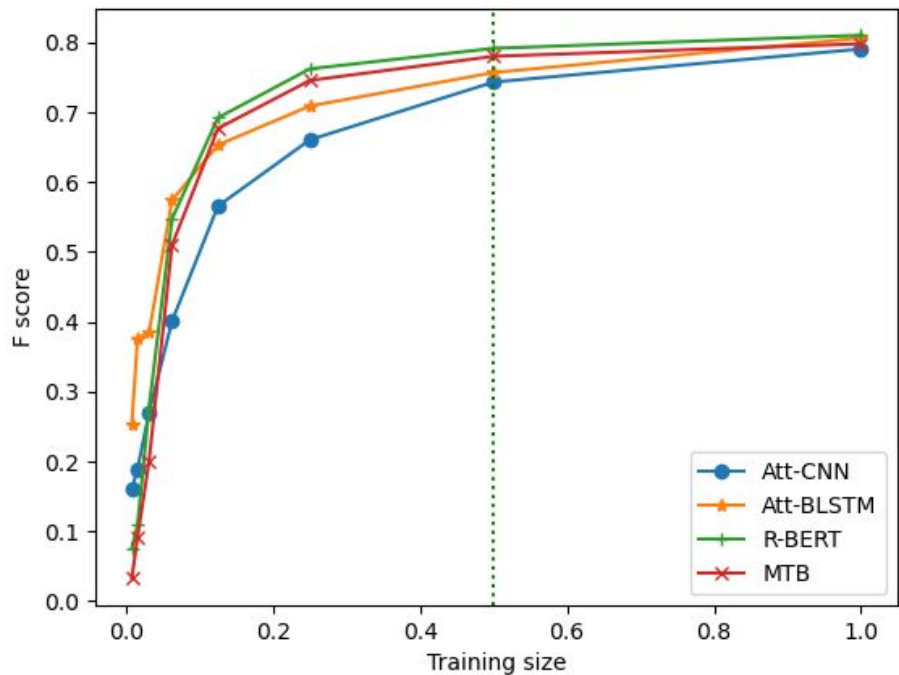
Transformer-based models (R-BERT, MTB) require less data to achieve strong results, making them more data-efficient compared to CNNs and BiLSTMs, likely due to pre-training.



Contribution - cont.

Findings

Simpler models like **Att-CNN** require larger datasets to reach competitive performance.



Conclusion

We have proposed how the size of a training dataset affects the performance of a relation classification task.

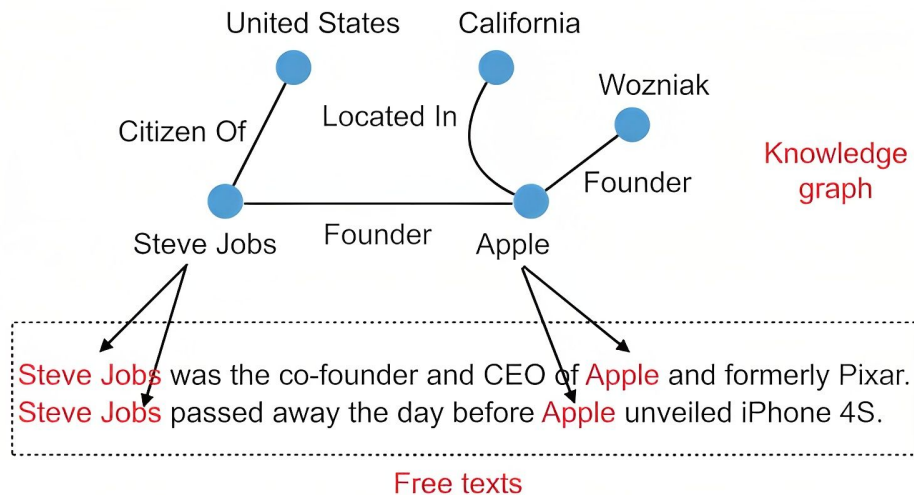
- Model performance improves with more instances
 - Feeding models additional data typically enhances their performance significantly.
- Diminishing returns with increased data
 - The benefits of increasing dataset size decline after reaching a certain point.
 - Strategies like data augmentation, active learning, or transfer learning may be more effective than simply increasing dataset size.

Thank you!
Any Questions
?

Appendix

Distant supervision methods (weakly supervised)

Aligning relational triplets in knowledge base with free texts to automatically generate labeled training dataset.

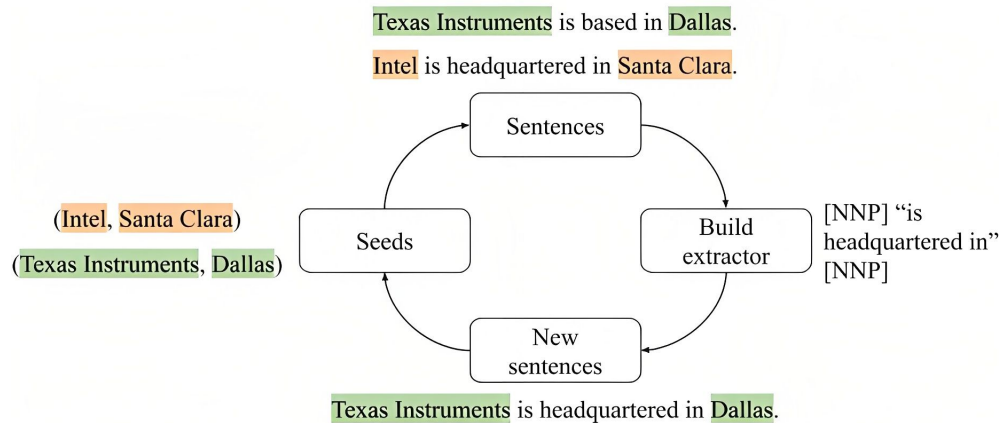


[Liu Kang - 2020]

Appendix

Semi-supervised methods

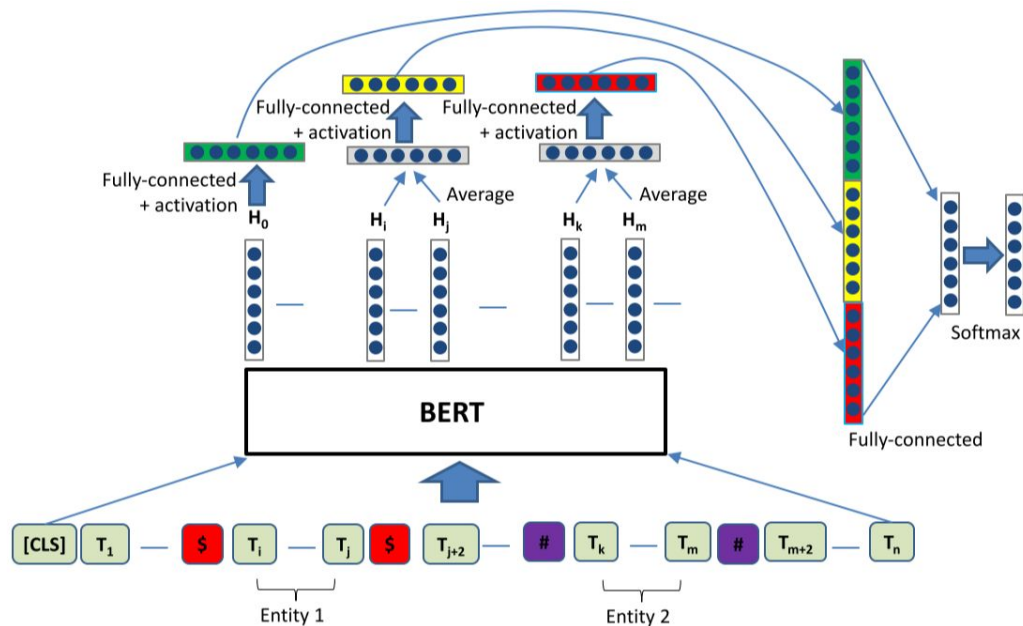
Methods in this setting typically rely on **bootstrapping**, or **bootstrap learning**.



[Liu Kang - 2020]

R-BERT - cont.

Model Architecture



[Wu and He - 2019]

$$H'_0 = W_0 (\tanh(H_0)) + b_0$$

$$H'_1 = W_1 \left[\tanh \left(\frac{1}{j-i+1} \sum_{t=i}^j H_t \right) \right] + b_1$$

$$H'_2 = W_2 \left[\tanh \left(\frac{1}{m-k+1} \sum_{t=k}^m H_t \right) \right] + b_2$$

$$h'' = W_3 \left[\text{concat} (H'_0, H'_1, H'_2) \right] + b_3$$

$$p = \text{softmax}(h'')$$

MTB - cont.

Pre-training Task

Key insight: Distributional similarity, Applied to relations.

$$h_r = f_{\Theta}(r)$$

[] forces had taken control of major cities in []'s east.

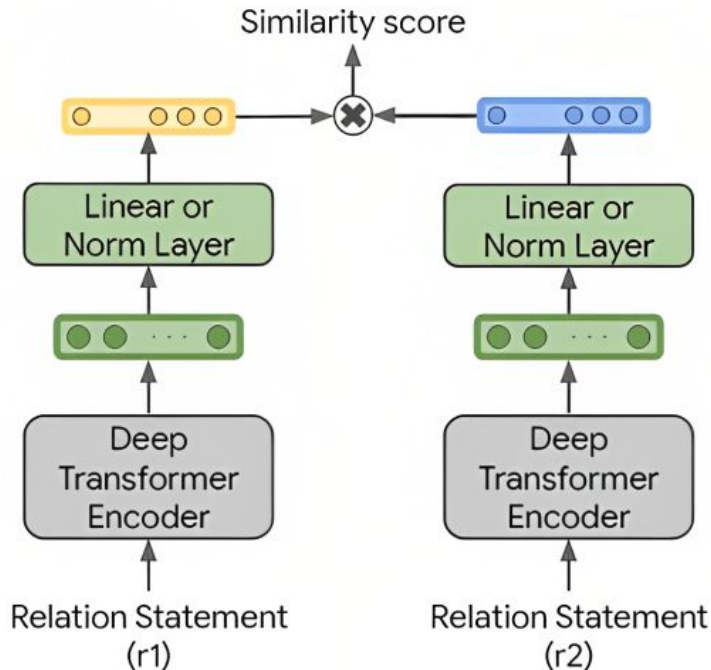
[] troops crossed []'s border bring with them hundreds of tanks.

The city's [] rulers and [] refugees may be long gone but their food has remained



MTB - cont.

Model Architecture

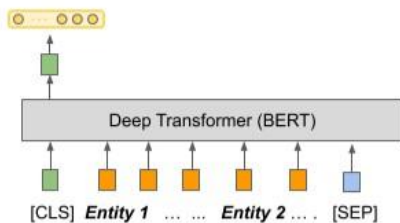


$$p(l = 1|\mathbf{r}, \mathbf{r}') = \frac{1}{1 + \exp f_{\theta}(\mathbf{r})^{\top} f_{\theta}(\mathbf{r}')}$$

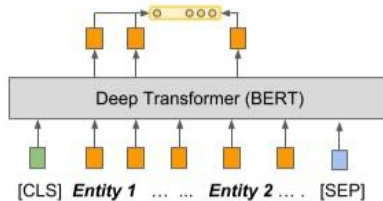
$$\mathcal{L}(\mathcal{D}) = -\frac{1}{|\mathcal{D}|^2} \sum_{(\mathbf{r}, e_1, e_2) \in \mathcal{D}} \sum_{(\mathbf{r}', e'_1, e'_2) \in \mathcal{D}} \delta_{e_1, e'_1} \delta_{e_2, e'_2} \cdot \log p(l = 1|\mathbf{r}, \mathbf{r}') + (1 - \delta_{e_1, e'_1} \delta_{e_2, e'_2}) \cdot \log(1 - p(l = 1|\mathbf{r}, \mathbf{r}'))$$

Appendix

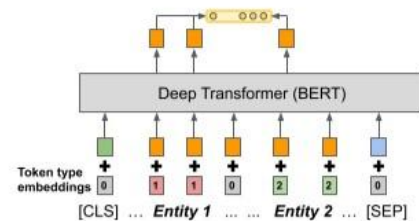
Input and output representation of MTB model



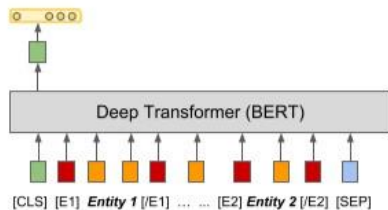
(a) STANDARD - [CLS]



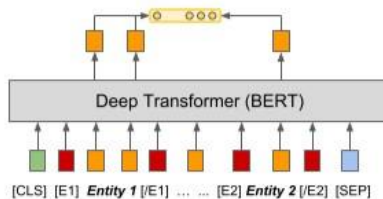
(b) STANDARD - MENTION POOLING



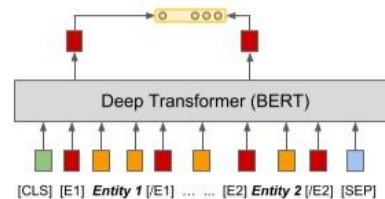
(c) POSITIONAL EMB. - MENTION POOL.



(d) ENTITY MARKERS - [CLS]



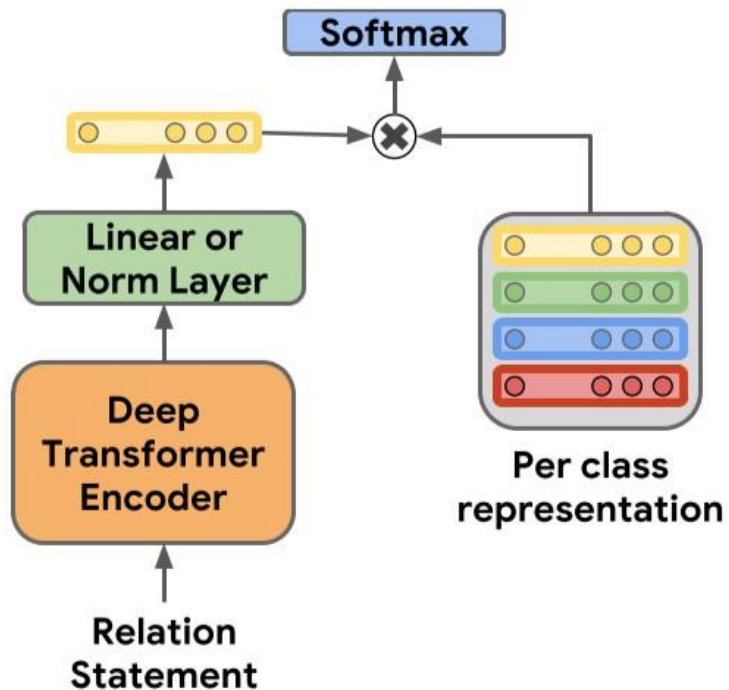
(e) ENTITY MARKERS - MENTION POOL.



(f) ENTITY MARKERS - ENTITY START

Appendix

MTB fine-tuning



Future Work

In the pre-training step of MTB model, we are going to ...

- Apply relation learning objective over larger corpus.
 - Number of semantically similar instances for each relation will increase.
- Utilize entity type feature of pre-training data.
 - Number of false positive instances will decrease.